



Sunbeam Institute of Information Technology

Pune and Karad

Module – Real Time Operating System

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Process/Thread $\rightarrow$ Task

Types of task :

a. Time driven task

1. Periodic task
2. Aperiodic task

b. Event driven task

3. Sporadic task (one shot task)

For each task :

- CPU burst (C_i)
- deadline (D_i)
- period (T_i)

1. Rate Monotonic Scheduling (RMS)
2. Deadline Monotonic Scheduling (DMS)
3. Earliest Deadline First (EDF) scheduling
4. Least slack Time First (LSTF) scheduling
Laziness

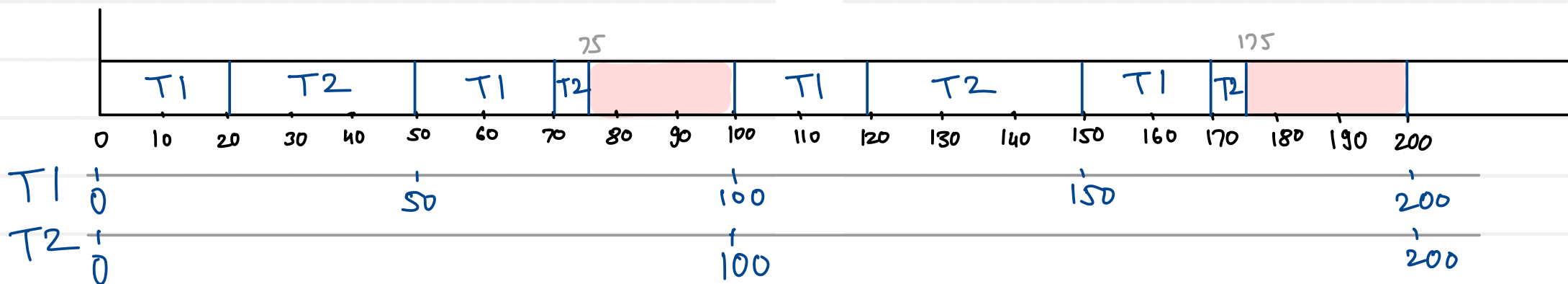
	CPU burst	Deadline	Period	Priority
T1	20	50	50	1
T2	35	100	100	2

Priorities: $T1 > T2$

Hyper period: $LCM(50, 100) = 100$

Release point / phase: 0

Priority $\propto \frac{1}{\text{period}}$



$$\text{CPU Utilization} = \frac{75}{100} = 0.75$$

↳ 75% CPU is busy

C_i = CPU time of task
 T_i = period of task

$$\text{CPU Utilization of task} = \frac{\text{CPU time}}{\text{period}} = \frac{C_i}{T_i}$$

$$\text{Total CPU Utilization} = \sum \text{CPU Utilization of tasks}$$

$$\text{Total CPU Utilization} = \frac{C_1}{T_1} + \frac{C_2}{T_2} + \dots$$

	CPU burst	Period
T1	20	50
T2	35	100

$$U_1 = \frac{C_1}{T_1} = \frac{20}{50} = 0.4$$

$$U_2 = \frac{C_2}{T_2} = \frac{35}{100} = 0.35$$

$$U = U_1 + U_2$$

$$= \frac{C_1}{T_1} + \frac{C_2}{T_2} = \frac{20}{50} + \frac{35}{100}$$

$$= 0.4 + 0.35$$

$$U = 0.75$$

RTOS scheduling - RMA

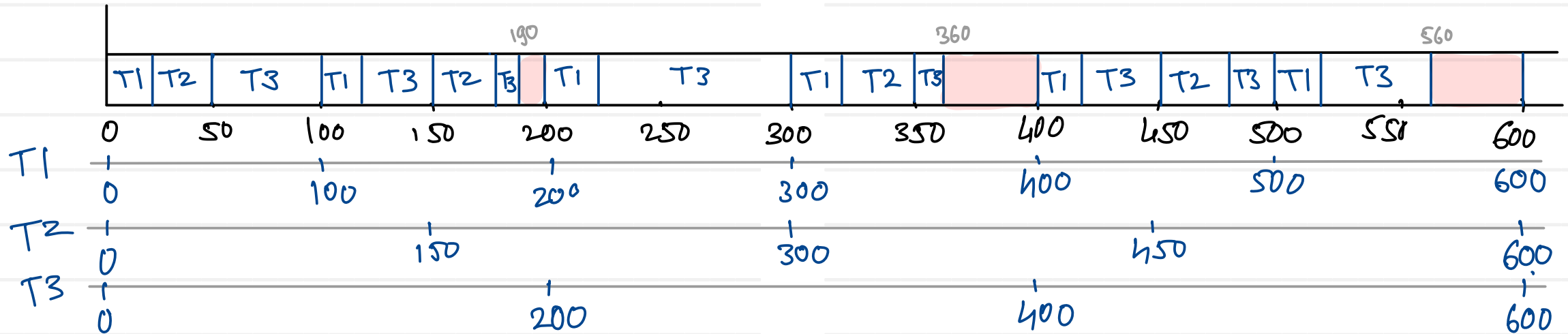
	CPU burst	Deadline	Period
T1	20	100	100
T2	30	150	150
T3	90	200	200

$$\text{CPU Utilization} = \frac{20}{100} + \frac{30}{150} + \frac{90}{200} = 0.85$$

Priority : $T1 > T2 > T3$

Hyper period = $\text{LCM}(100, 150, 200) = 600$

phase = 0



$$\text{idle time} = 10 + 40 + 40 = 90$$

$$\text{CPU Utilization} = \frac{510}{600} = 0.85$$

	CPU burst	Deadline	Period	Priority
T1	3	7	20	2
T2	2	4	5	1
T3	2	9	10	3

$$\text{Priority} \propto \frac{1}{\text{deadline}}$$

if deadline is not given, period is considered as deadline. It works same as RMS

$$\text{CPU Utilization} = \frac{3}{20} + \frac{2}{5} + \frac{2}{10} = 0.75$$

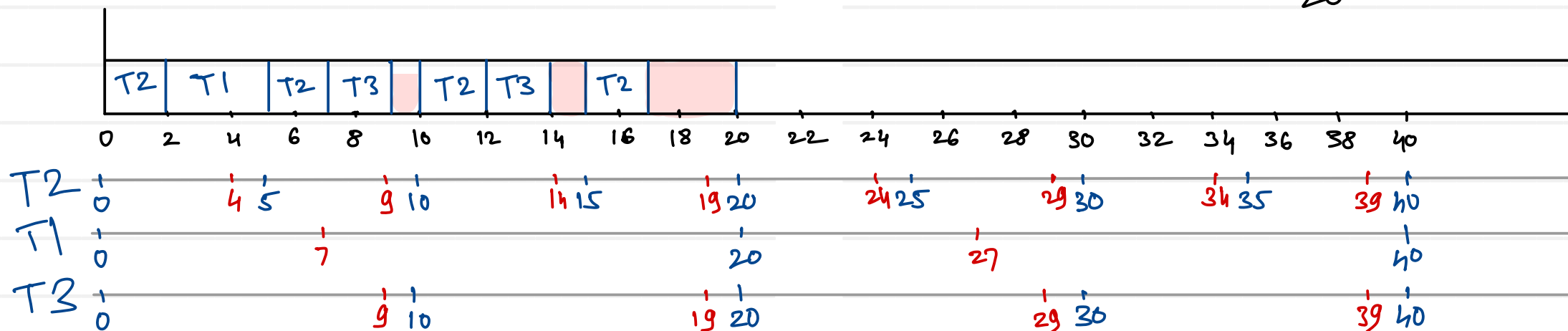
$$\text{Hyper period} = \text{LCM}(20, 5, 10) = 20$$

$$\text{Priorities} = T2 > T1 > T3$$

$$\text{Release point} = 0$$

$$\text{idle time} = 1 + 1 + 3 = 5$$

$$\text{CPU Utilization} = \frac{15}{20} = 0.75$$



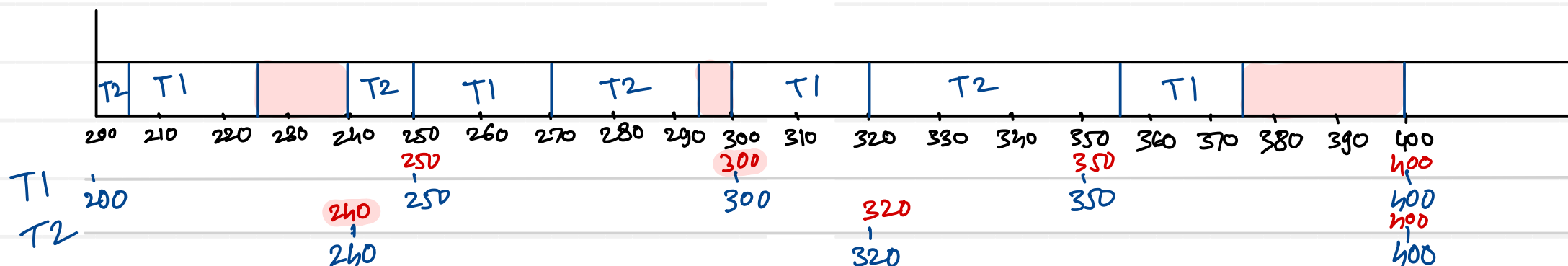
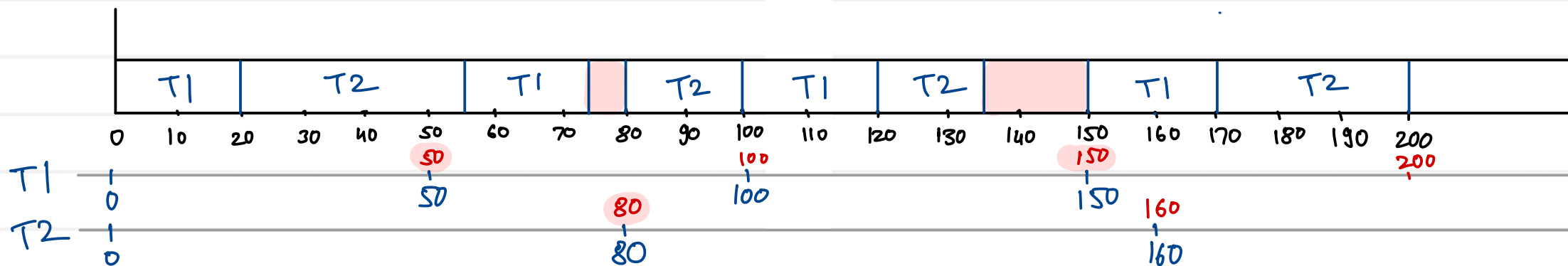
	CPU burst	Deadline	Period
T1	20	50	50
T2	35	80	80

priority $\propto$ earliest deadline

$$\text{CPU Utilization} = \frac{20}{50} + \frac{35}{80} = 0.83$$

$$\text{Hyper period} = \text{LCM}(50, 80) = 400$$

Release point = 0





Thank you!!!

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